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Surgical guide and fixation insertion device

Abstract

A surgical guidance device is cannulated for passage of a guidewire or pin and includes a hand grip and differential control assembly for driving the guidewire or pin and differentially driving a screw, such as but not limited to a pedicle screw, into bone. The differential control assembly resides in a housing, and includes a differential control assembly for gripping and locking or releasing the guidewire or pin and effecting directional movement (relative to the user holding the surgical guidance device) either distally into the bone or proximally away from bone to thereby drive or retract the guidewire. The differential control assembly also allows distal or proximal drive of a screw, such that one or both the guidewire and screw can be passed proximally or distally alone or together.

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Background/Summary

RELATED APPLICATIONS (1) This application claims the benefit of and priority to U.S. provisional application No. 63/428,713 filed Nov. 29, 2022, and U.S. provisional application No. 63/330,224 filed Apr. 12, 2022, which are hereby incorporated herein by reference in their entireties.

FIELD OF THE INVENTION

(1) The invention relates to a surgical guidance device and system and a method for use of the surgical guidance device for surgical spinal procedures. In particular, the surgical guidance device includes a driving tool and differential drive system for inserting and positioning, sequentially, a guidewire and a screw into bone, in particular for inserting a cannulated pedicle screw into a

vertebra over a guidewire.

BACKGROUND OF THE INVENTION

(2) Medical procedures calling for the placement of a screw or other fixation implant within bone often rely upon initial placement of a guide, such as a pin or a guidewire, over which the screw or implant is guided for insertion within bone. In some examples, medical procedures involving the vertebrae can be particularly complicated because of the precision and accuracy required to avoid both neural damage and injury to major blood vessels when placing the guide and then driving the implant into bone. Precision depth guided devices may be employed for such applications, in particular when the mode of surgical access is percutaneous. In the context of spine surgeries, the surgical technique typically involves penetration of the hard cortical bone of the vertebra then traversal of the softer cancellous bone lying thereunder. A concomitantly greater force is typically required to penetrate the cortical bone as compared to the force needed for passage through the underlying cancellous bone. Careful control of the forces and depth of penetration of the guide and the implant are essential to avoid damage to the spinal cord and/or other organs or blood vessels located adjacent the spine. Current solutions for placement of guidewires and pins and subsequent placement of screws rely on various manually or power driven penetration which suffer from limited control to avoid tissue damage.

(3) There remains a need in the art for a device and system and method that enables precise and differentially controlled insertion of the guidewire and screw, and in particular for precisely controlling the depth that the guidewire can penetrate. The guidance device, system and method should also be constructed to allow the user to precisely retract the guidewire through bone as needed.

SUMMARY

(4) In various embodiments, the disclosure provides a surgical guidance device **10**, system and method for forming a pilot bore and passing a guidewire (or pin) **15** into bone based on a trajectory established using radiography (for example, but not limited to, 2D or 3D fluoroscopy or CT).

(5) The surgical guidance device **10** is cannulated for passage of a guidewire or pin and includes a hand grip **40** and differential control assembly **50** for driving the guidewire **15** (or pin) and differentially driving a screw, such as but not limited to a pedicle screw, into bone. The differential control assembly **50** resides in a housing **20**, and includes a differential control assembly **50** for gripping and locking or releasing the guidewire **15** and effecting directional movement (relative to the user holding the surgical guidance device **10**) either distally into the bone or proximally away from bone to thereby drive or retract the guidewire. The differential control assembly **50** also allows distal or proximal drive of a screw, such that one or both the guidewire and screw can be passed proximally or distally alone or together.

(6) One key functional feature of the surgical guidance device **10** is the ability to establish a preselected depth limit on the passage of the guidewire or pin whereupon the surgical guidance device **10** can be activated passively by means of a positive stop feature, or by user actuation to lock the guidewire or pin and prevent further distal advancement of the guidewire **15** or pin, and the device can be further actuated to actively withdraw the guidewire **15** and isolate its movement while directing movement of only a screw. Accordingly, the device prevents passage of a guidewire or pin farther into bone than the preselected depth limit and operates to withdraw the guidewire or pin without affecting the placement of a screw.

(7) In an exemplary embodiment, the surgical guidance device **10** includes a differential control assembly **50** that is cannulated for passage of a guidewire **15** therethrough, and aligned along a common axis with and within a housing **20**, the differential control assembly **50** including a support shaft **51** having a proximally positioned drive connector **59** for engagement with a powered driving device, and a distally positioned tool connector **52** configured for attachment of a driver tool guide **200**, a screw gear **53** and a chuck **54** that are releasably engagable to direct rotational movement of the screw gear **53** when the chuck **54** is engaged therewith, and a threaded carriage **55** translatable

along the support shaft **51** for directing proximal or distal movement of the guidewire **15** through the differential control assembly **50**, the threaded carriage **55** including at least one positive stop **57** which limits the proximal and distal translation of the carriage **55** within the housing **20**, and an actuator assembly **29** including a handle **42** operably attached to a chuck mover **43** for actuating engagement of the chuck **54** with the screw gear **53**, a directional actuator **30** for actuating movement of the carriage **55** proximally or distally, and a carriage lock **31** configured to releasably engage a thread guide **32** with the threaded carriage **55** wherein engagement thereof permits proximal or distal movement of the threaded carriage **55** along the support shaft **51** to effect proximal or distal movement of the guidewire **15**, wherein the differential control assembly **50** is actuatable to first drive the guidewire **15** into a target site by engagement of the carriage lock **31** with the carriage **55** to direct distal movement of the guidewire **15**, the depth thereof being governed by the at least one positive stop **57**, and to differentially drive a screw affixed to the surgical guidance device **10** by actuating the handle **42** to engage the chuck **54** with the screw gear **53**.

(8) In an embodiment, the surgical guidance device **10** includes a housing **20** having a proximal end **21**, a center drive barrel **24**, and a distal end **22**, the proximal end **21** adapted with a hand grip **40** for gripping the surgical guidance device **10**, the center drive barrel **24** configured to contain a differential control assembly **50** housed therein, the proximal end **21** of the housing **20** adapted for securing the housing to a powered driving device such as a drill or power drill, the differential control assembly **50** being cannulated for passage of a guidewire therethrough, and aligned along a common axis with and within the housing **20**, the differential control assembly **50** including a support shaft **51** having a proximally positioned drive connector **59** for engagement with the powered driving device, and a distally positioned tool connector **52** configured for attachment of a driver tool guide **200**, a screw gear **53** and a chuck **54** that are releasably engagable to direct rotational movement of the screw gear **53** when the chuck **54** is engaged therewith, a threaded carriage **55** translatable along the support shaft **51** for directing proximal or distal movement of the guidewire **15** through the differential control assembly **50**, the threaded carriage **55** including at least one positive stop **57** which limits the proximal and distal translation of the carriage **55** within the housing **20**, and an actuator assembly **29** including a hand grip **40** including a handle **42** operably attached to a chuck mover **43** for actuating engagement of the chuck **54** with the screw gear **53**, a directional actuator **30** for actuating movement of the carriage **55** proximally or distally, and a carriage lock **31** configured to releasably engage a thread guide **32** with the threaded carriage **55** wherein engagement thereof permits proximal or distal movement of the threaded carriage **55** along the support shaft **51** to effect proximal or distal movement of the guidewire **15**, wherein the differential control assembly **50** is actuatable to first drive the guidewire **15** into a target site by engagement of the carriage lock **31** with the carriage **55** to direct distal movement of the guidewire **15**, the depth thereof being governed by the at least one positive stop **57**, and to differentially drive a screw affixed to the surgical guidance device **10** by actuating the handle **42** to engage the chuck **54** with the screw gear **53**.

(9) In some embodiments, the surgical guidance device **10** includes a fixed grip **41**.

(10) In some embodiments, the driver tool **200** includes a drill guide.

(11) In some embodiments, the differential control assembly **50** includes a connector **51** for securing to a powered driving device such as a drill or power drill.

(12) In some embodiments, the surgical guidance device **10** includes a trigger **70** for disengaging the screw gear **53** to stop rotation of a screw affixed to the surgical guidance device **10**.

(13) In some embodiments, the surgical guidance device **10** includes a guidewire lock **71** for locking the guidewire in place to prevent one or more of its rotation, proximal translation, distal translation, or a combination thereof.

(14) In some embodiments, the driver tool **200** is constructed to cooperate with a conjugately shaped portion of a screw and is adapted to receive passage of the guidewire therethrough.

- (15) In some embodiments, the directional actuator **30** is actuated to direct the guidewire **15** to move distally, and is released to retract the guidewire to move proximally, wherein displacement of the guidewire **15** is governed by the at least one positive stop **57**, and wherein retraction of the guidewire **15** with the directional actuator **30** thereby differently permits distally directed driving of only the screw gear **53**.
- (16) In some embodiments, the guidewire **15** is constructed from a material having sufficient column strength for penetrating a bone.
- (17) In some embodiments, the surgical guidance device **10** has a fixed length from proximal to distal wherein rotation and translation of the differential control assembly **50** occurs within the housing **20**.
- (18) In some embodiments, the proximal end of the housing **20** is integral with a powered driving device **500**.
- (19) In some embodiments, the at least one positive stop is at least one ring that interferes within the housing to limit movement of the carriage **55**.
- (20) In some embodiments, at least one positive stop includes two or more rings oriented proximally and distally relative to the carriage **55**.
- (21) In some embodiments, the positive stop is a sleeve that encloses the carriage **55** to thereby limit its translation.
- (22) In an embodiment, the invention includes a method for directing sequentially a guidewire **15** and a screw into target bone by differentially directing the placement of a guidewire into bone and thereupon preventing further displacement of the guidewire **15** into bone and thereafter introducing the screw into bone, wherein the guidewire **15** is either maintained in place within the bone, or is withdrawn before or as the screw is inserted into the bone.
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Description

DESCRIPTION OF THE DRAWINGS

- (1) The features and advantages of the general inventive concepts will become apparent from the accompanying drawings which depict various alternate embodiments of the surgical guidance device and method. The drawings illustrate various embodiments of the present invention and methods by which they may be assembled and operated. Like reference numerals refer to like components in the various figures.
- (2) FIG. **1** is a side perspective view schematic of a first embodiment of the surgical guidance device according to the disclosure;
- (3) FIG. **2** is partial side perspective view schematic of the embodiment of the surgical guidance device shown in FIG. **1** wherein the surgical guidance device is shown with a guidewire passed through its central cannulation and in the included differential control assembly embodiment thereof is in a neutral position;
- (4) FIG. **3** is partial side perspective view schematic of the embodiment of the surgical guidance device shown in FIG. **1**-FIG. **2** wherein the surgical guidance device is shown with a guidewire passed through its central cannulation and retracted toward the proximal (right side) end of the surgical guidance device;
- (5) FIG. **4** is partial side perspective view schematic of the embodiment of the surgical guidance device shown in FIG. **1**-FIG. **3** wherein each of the chuck mover and the directional actuator are actuated to direct distal movement of the screw gear and proximal movement of the guidewire;
- (6) FIG. **5** is partial side perspective view schematic of the embodiment of the surgical guidance device shown in FIG. **1**-FIG. **4** wherein the guidewire is locked;
- (7) FIG. **6** is partial side perspective view schematic of the embodiment of the surgical guidance device shown in FIG. **1**-FIG. **5** wherein the threaded carriage is engaged prior to actuation of the

carriage lock;

(8) FIG. 7 is a close-up, side perspective view schematic of the carriage lock mechanism as shown in FIG. 1-FIG. 6;

(9) FIG. 8 is a partial side perspective view schematic of the embodiment of the surgical guidance device shown in FIG. 1-FIG. 7 wherein the threaded carriage is disengaged by actuation of the carriage lock;

(10) FIG. 9 is a schematic showing in a vertical orientation a side view portion of an embodiment of a differential control assembly of the surgical guidance device, the depicted differential control assembly connectable at its proximal (top) end to a drill, including a screwdriver sleeve with locking pin, and a carriage including a driving chuck, with pin notches region and stops and a washer to limit proximal to distal motion the assembly having a guidewire passed through;

(11) FIG. 10 is a schematic showing in a vertical orientation a driving chuck portion of a differential control assembly with a guidewire passed through and a locking tray to lock rotation of the driving chuck;

(12) FIG. 11 is a schematic showing a guidance device including a proximal handle (right) engaged or engageable with a drill, a trigger connected to a differential control assembly as shown in FIG. 9 and FIG. 10, and a distal driver in a vertical orientation a portion of a differential control assembly with a guidewire passed through;

(13) FIG. 12 is a schematic showing in two views the guidance device shown in FIG. 11 wherein movement of the trigger from distal to proximal directs engagement of the driving chuck to permit driving of one of a screw or a guidewire;

(14) FIG. 13 is a top view of an embodiment of the washer having interior threads for engaging a threaded portion of the carriage;

(15) FIG. 14 is a side perspective schematic view of a second embodiment of the surgical guidance device according to the disclosure;

(16) FIG. 15 includes a photographic image of a physical embodiment of the depicting the surgical guidance device shown in FIG. 14, and a partial side perspective view schematic of the embodiment of the surgical guidance device shown in FIG. 14 wherein the surgical guidance device is shown with a guidewire passed through its central cannulation;

(17) FIG. 16 is schematic showing sequential images of the embodiment of the surgical guidance device shown in FIG. 14-15, each image including corresponding flow diagram descriptions of the actuation of features of the device;

(18) FIG. 17 is a side perspective schematic view and a side schematic view of a third embodiment of the surgical guidance device according to the disclosure;

(19) FIG. 18 is schematic showing sequential images of the embodiment of the surgical guidance device shown in FIG. 17, each image including corresponding flow diagram descriptions of the actuation of features of the device

(20) FIG. 19 is a side perspective schematic view of a fourth embodiment of the surgical guidance device according to the disclosure; and

(21) FIG. 20 is a side perspective schematic view of a fifth embodiment of the surgical guidance device according to the disclosure.

(22) This disclosure describes exemplary embodiments in accordance with the general inventive concepts and is not intended to limit the scope of the invention in any way. Indeed, the invention as described in the specification is broader than and unlimited by the exemplary embodiments set forth herein, and the terms used herein have their full ordinary meaning.

DESCRIPTION

(23) In various embodiments, the invention provides an integrated, all-in-one powered device that can deploy a guidewire or pin, for example a surgical K-wire, or a screw or other fixator, such as a pedicle screw, or both. The surgical guidance device 10 is adapted to be attached to a conventional drill, or may be integrated into a drill device, or may have integrated onboard power for driving, as

shown in alternate embodiments herein.

(24) The surgical guidance device **10** is cannulated for passage of a guidewire or pin and includes a hand grip **40** and differential control assembly **50** for driving the guidewire **15** or pin and differentially driving a screw, such as but not limited to a pedicle screw, into bone. The differential control assembly **50** resides in a housing **20**, and includes a differential control assembly **50** for gripping and locking or releasing the guidewire or pin and effecting directional movement (relative to the user holding the surgical guidance device **10**) either distally guidewire or pin into the bone or proximally away from bone to thereby drive or retract the guidewire. The differential control assembly **50** also allows distal or proximal drive of a screw, such that one or both the guidewire and screw can be passed proximally or distally alone or together.

(25) One key functional feature of the surgical guidance device **10** is the ability to establish a preselected depth limit on the passage of the guidewire or pin whereupon the surgical guidance device **10** can be activated passively by means of a positive stop feature, or by user actuation to lock the guidewire or pin and prevent further distal advancement of the guidewire **15** or pin, and the device can be further actuated to actively withdraw the guidewire **15** or pin and isolate its movement while directing movement of only a screw. Accordingly, the device prevents passage of a guidewire or pin farther into bone than the preselected depth limit and operates to withdraw the guidewire or pin without affecting the placement of a screw.

(26) In an exemplary embodiment, the surgical guidance device **10** includes a differential control assembly **50** that is cannulated for passage of a guidewire **15** therethrough, and aligned along a common axis with and within a housing **20**, the differential control assembly **50** including a support shaft **51** having a proximally positioned drive connector **59** for engagement with a powered driving device, and a distally positioned tool connector **52** configured for attachment of a driver tool guide **200**, a screw gear **53** and a chuck **54** that are releasably engagable to direct rotational movement of the screw gear **53** when the chuck **54** is engaged therewith, and a threaded carriage **55** translatable along the support shaft **51** for directing proximal or distal movement of the guidewire **15** through the differential control assembly **50**, the threaded carriage **55** including at least one positive stop **57** which limits the proximal and distal translation of the carriage **55** within the housing **20**, and an actuator assembly **29** including a handle **42** operably attached to a chuck mover **43** for actuating engagement of the chuck **54** with the screw gear **53**, a directional actuator **30** for actuating movement of the carriage **55** proximally or distally, and a carriage lock **31** configured to releasably engage a thread guide **32** with the threaded carriage **55** wherein engagement thereof permits proximal or distal movement of the threaded carriage **55** along the support shaft **51** to effect proximal or distal movement of the guidewire **15**. The differential control assembly **50** is actuatable to first drive the guidewire **15** into a target site by engagement of the carriage lock **31** with the carriage **55** to direct distal movement of the guidewire **15**, the depth thereof being governed by the at least one positive stop **57**, and to differentially drive a screw affixed to the surgical guidance device **10** by actuating the handle **42** to engage the chuck **54** with the screw gear **53**.

(27) The guidewire can be a surgical K-wire or a pin or other internal guiding pin like device. Power may direct pin motion in a distal or proximal direction by rotating or malleating and would rotate the screw upon engagement of the chuck **54** with the screw gear **53**. In some embodiments, the surgical guidance device **10** is particularly adapted to differentially drive the guidewire, the screw or both, and may drive the screw distally while withdrawing the guidewire proximally.

(28) Thus, in some embodiments, the surgical guidance device **10** drives only the guidewire and holds the screw static, or drives only the screw holding the guidewire static, or drives the screw either proximally or distally while withdrawing the guidewire proximally (i.e., withdrawing from the bone). The invention is thus advantageous as compared to existing solutions which rely on separately actuated pin and screw advancement/withdrawal, and further may rely solely on manual driving without power.

(29) Referring now to the drawings as variously depicted in FIG. 1-FIG. 20, various embodiments

and configurations of the surgical guidance device **10** are shown.

(30) Referring now to FIGS. **1-8**, an embodiment of the surgical guidance device **10** is shown in which the surgical guidance device **10** is shown in FIG. **2-8** in series from initial assembly with the guidewire **15** passed through the cannula in the differential control assembly **50**, through to engagement driver tool guide **200** (tool not shown), for example, in FIG. **8**, wherein an housing **20** is shown as transparent, and the internal differential control assembly **50** are shown through the transparent housing **20**. As shown in FIG. **8**, within the housing are a carriage with stops/sleeve, chuck for driving the screwdriver, and screw gear components.

(31) Referring again to the drawings as specifically depicted in FIG. **9-FIG. 13**, an embodiment of a differential control assembly **50** is shown in which the guidewire **15** is advanced by actuation of the surgical guidance device **10** while the screwdriver component is disengaged from power. In a representative example, the guidewire **15** may be advanced by a fixed distance, for example for total length of 10 mm upon reaching this distance, the carriage hit a stop. Upon reaching the stop, an actuator, such as a button, as shown in FIG. **10**, is actuated to allow the drill to engage to start driving the screw. Referring to FIG. **12**, an optional locking button is a tray is introduced from below.

(32) In various embodiments, it will be appreciated that any predetermined depth of travel of the guidewire, forward or backward, alone or together with the screw, may be selected from 0-50 mm, or from 5-30 mm, or from 10-20 mm, or 15 mm, or any other range or increment.

(33) Referring again to the drawings as specifically depicted in FIG. **14-FIG. 15**, the surgical guidance device **10** is shown, wherein an outer housing is shown as transparent, and the internal differential control assembly **50** are shown through the transparent housing **20**. In this embodiment, a locking button **71** for securing and thereby preventing movement of the guidewire is positioned at the proximal end of the housing/carriage.

(34) In some embodiments, the surgical guidance device **10** will either have a fixed stop to prevent translation of the carriage or the carriage or the stop is adjustable to allow for adjustable guidewire displacement.

(35) In some embodiments, the carriage will freely float proximal and distal along the differential control assembly **50** lacking the at least one stop.

(36) Referring again to FIG. **9-13**, in some embodiments, such as shown in FIG. **9-13**, the stop may be an adjustably positionable washer, or a locking pin for engaging with locking pin notches, or a sleeve.

(37) Referring again to FIG. **9-13**, in some embodiments, the carriage can be rotationally locked into the screwdriver by engaging the locking pin or pins into the locking grooves. In some embodiments, the grooves allow for the carriage to displace rotational power, but still allow for proximal and distal translation along the differential control assembly **50**. In some embodiments, the distraction washer is adjustable and stops the depth the guidewire can be driven. When the washer is rotated it can either permit a greater length of guidewire to advance distally, or retract the guidewire. In some embodiments, the distraction washer has a reverse thread, so after the locking button is pushed, the user's hand is removed from the screwdriver and placed on the distraction washer. In some embodiments, when the drill is turned on the screw will drive and the carriage will be pulled back in relation to the screwdriver. Thus, in such embodiments, the effect is to retract the guidewire **15**.

(38) More generally, in those embodiments including a sleeve or stop, the surgical guidance device **10** is operated such that the sleeve or stop interacts with or without a distraction washer. In the instance of a distraction washer, the washer contacting the sleeve is forced to spin towards the screwdriver. Once the washer hits the screwdriver it will back up the guidewire **15**. To avoid premature withdrawal of the guidewire, a selected length of tread, for example, 10 mm ensures that if the screw isn't advanced into the bone upon engagement of the driver, the guidewire **15** is maintained in place until the screw is advanced the selected distance. The handle as controlled by

the user to maneuver the sleeve to interact with the washer at the same time the lock is engaged with the screwdriver handle.

(39) In various embodiments, in use, the distal end of the surgical guidance device **10** is docked, with the screw in direct contact on bone, for example in a representative spinal surgery, on the Pars interarticularis, which is the small segment of bone that joins the facet joints in the back of the spine.

(40) Upon engagement of the screwdriver, the screw is driven, for example for a distance of about 10 mm. In some embodiments the guidewire **15** length maintained at 10 mm. Preservation of the guidewire **15** placement in bone is beneficial in the instance where initial purchase of the screw tip in bone is not established, thus ensuring that the trajectory is not lost. In other conventional approaches, early withdrawal of the guidewire **15** prior to engagement of the screw in bone can require extensive additional steps including additional radiographic shots in order to reestablish wire placement in bone. As the screw is driven further into the bone, the guidewire **15** may be retracted either by automated actuation or by user actuation. For example, the locking button is depressed releasing the guidewire **15** to enable its retraction. Thus, the further additional depth screw is driven, the guidewire **15** starts retracting, in some examples, within 10 additional mm.

(41) It will be appreciated that while the embodiments herein include a differential control assembly **50** having various differences in their components, the embodiments may be modified to include any of the differential control assembly **50** components and actuators as described herein. For example, some or all of the features of the embodiment of a differential control assembly **50** as specifically depicted in FIG. 9-FIG. 13 may be incorporated in a surgical guidance device **10** according to the disclosure.

(42) Referring again to the drawings, FIG. 14 shows an embodiment of the surgical guidance device **10** which differs in some respects from the embodiment shown in FIG. 1-8, and FIG. 15 shows the embodiment of FIG. 14 including a photographic image of a prototype of the surgical guidance device **10** showing engagement with a drill on the proximal end, and at the distal end, a pedicle screw with affixed tulip head is attached to the distal end of the driver via a locking nut, and the drive is in turn affixed to the surgical guidance device **10** above the locking nut. FIG. 15 also shows a partial view of the surgical guidance device **10** with transparent rendering of the housing **20** to show the differential control assembly **50**.

(43) Referring now to FIG. 16 a schematic shows sequential images of the embodiment of the surgical guidance device shown in FIG. 14-15, each image including corresponding flow diagram descriptions of the actuation of features of the device. FIG. 16 provides a stepwise guide to actuation of the surgical guidance device **10** consistent with the above described methods of its use, whereby the surgical guidance device **10** under power, for example from a conventional drill, enables insertion of a guidewire followed by insertion of a screw and removal of the guidewire without the need for any other devices.

(44) Referring again to the drawings, FIG. 17-FIG. 18 shows an alternate embodiment of the surgical guidance device **10** which includes a fully integrated power drill that enables single handed control. FIG. 18 is schematic showing sequential images of the embodiment of the surgical guidance device shown in FIG. 17, each image including corresponding flow diagram descriptions of the actuation of features of the device, and provides a stepwise guide to actuation of the surgical guidance device **10** consistent with the above described methods of its use, whereby the surgical guidance device **10** under power, for example from a conventional drill, enables insertion of a guidewire followed by insertion of a screw and removal of the guidewire without the need for any other devices.

(45) The embodiment shown in FIG. 1-FIG. 23 can be used by a single operator using two hands, one to control and actuate the surgical guidance device **10**, and one to actuate the drill.

(46) Referring again to the drawings, FIG. 19 shows yet another embodiment of the surgical guidance device **10** with further streamlined controls for single handed use, and adapted with a

push button mechanism directional actuator **30** for retraction of the guidewire once acceptable screw depth has been achieved.

(47) And FIG. **20** shows yet another embodiment of the surgical guidance device **10**, wherein the device includes a simple push driver button **72** at its distal end to drive insertion of a guidewire.

(48) In some embodiments, the present device **10** may be utilized for any of a variety of surgical procedures where precision of placement and control of at least a guidewire with or without a screw is needed, for example in soft tissue procedures for placement of instruments and or biopsies or removal of tissues, and in orthopedic procedures outside the spine for example in long bones, and joints including the shoulder, hip, knee or other joint.

PARTS LIST

(49) surgical guidance device **10** differential control assembly **50** guide wire **15** housing **20** support shaft **51** proximally positioned drive connector **59** distally positioned tool connector **52** driver tool guide **200** screw gear **53** chuck **54** threaded carriage **55** at least one positive stop actuator assembly **29** handle **42** chuck mover **43** directional actuator **30** carriage lock **31** thread guide **32** fixed grip **41**. trigger **70** wire lock **71** push driver button **72** screw **300** power drill **500**

(50) This description describes exemplary embodiments in accordance with the general inventive concepts and is not intended to limit the scope of the invention in any way. Indeed, the invention as described in the specification is broader than and unlimited by the exemplary embodiments set forth herein, and the terms used herein have their full ordinary meaning.

(51) The general inventive concepts will now be described with occasional reference to the exemplary embodiments of the invention. Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art encompassing the general inventive concepts. The terminology set forth in this detailed description is for describing particular embodiments only and is not intended to be limiting of the general inventive concepts.

(52) As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise.

(53) Unless otherwise indicated, all numbers expressing quantities, properties, and so forth as used in the specification and claims are to be understood as being modified in all instances by the term “about.” Accordingly, unless otherwise indicated, the numerical properties set forth in the specification and claims are approximations that may vary depending on the suitable properties desired in embodiments of the present invention. Notwithstanding that the numerical ranges and parameters setting forth the broad scope of the general inventive concepts are approximations, the numerical values set forth in the specific examples are reported as precisely as possible. Any numerical values, however, inherently contain certain errors necessarily resulting from error found in their respective measurements.

(54) The term “proximal” as used in connection with any object refers to the portion of the object that is closest to the operator of the object (or some other stated reference point), and the term “distal” refers to the portion of the object that is farthest from the operator of the object (or some other stated reference point). The term “operator” means and refers to any professional or paraprofessional who delivers clinical care to a medical patient, particularly in connection with the delivery of care. A “clinical subject” refers to a human or other animal who is the subject of treatment with a bone fixation or reduction device in accordance with the disclosure. With respect to any references herein that may be made relative to a clinical subject, the term “medial” indicates a direction toward the centerline axis (e.g., the spine) of the clinical subject, and the term “lateral” indicates a direction toward a side of the clinical subject. The term “system” refers to any combination of two or more of objects, devices, or components. And the term “method” refers to any one of methods of using a device or system, and surgical methods or techniques employing a device or system.

(55) While various inventive aspects, concepts, and features of the general inventive concepts are

described and illustrated herein in the context of various exemplary embodiments, these various aspects, concepts, and features may be used in many alternative embodiments, either individually or in various combinations and sub-combinations thereof. Unless expressly excluded herein, all such combinations and sub-combinations are intended to be within the scope of the general inventive concepts. Still further, while various alternative embodiments as to the various aspects, concepts, and features of the inventions (such as alternative materials, structures, configurations, methods, devices and components, alternatives as to form, fit and function, and so on) may be described herein, such descriptions are not intended to be a complete or exhaustive list of available alternative embodiments, whether presently known or later developed.

(56) Those skilled in the art may readily adopt one or more of the inventive aspects, concepts, or features into additional embodiments and uses within the scope of the general inventive concepts even if such embodiments are not expressly disclosed herein. Additionally, even though some features, concepts or aspects of the inventions may be described herein as being a preferred arrangement or method, such description is not intended to suggest that such feature is required or necessary unless expressly so stated. Still further, exemplary, or representative values and ranges may be included to assist in understanding the present disclosure; however, such values and ranges are not to be construed in a limiting sense and are intended to be critical values or ranges only if so expressly stated. Moreover, while various aspects, features and concepts may be expressly identified herein as being inventive or forming part of an invention, such identification is not intended to be exclusive, but rather there may be inventive aspects, concepts and features that are fully described herein without being expressly identified as such or as part of a specific invention. Descriptions of exemplary methods or processes are not limited to inclusion of all steps as being required in all cases, nor is the order that the steps are presented to be construed as required or necessary unless expressly so stated. Further, while disclosed benefits, advantages, and solutions to problems have been described with reference to specific embodiments, these are not intended to be construed as essential or necessary to the invention.

Claims

1. A surgical guidance device (10) that comprises: a. a differential control assembly (50) that is cannulated for passage of a guidewire (15) therethrough, and aligned along a common axis with and within a housing (20), the differential control assembly (50) including a support shaft (51) having a proximally positioned drive connector (59) for engagement with a powered driving device (500), and a distally positioned tool connector (52) configured for attachment of a driver tool guide (200), a screw gear (53) and a chuck (54) that are releasably engagable to direct rotational movement of the screw gear (53) when the chuck (54) is engaged therewith, and a threaded carriage (55) translatable along the support shaft (51) for directing proximal or distal movement of the guidewire (15) through the differential control assembly (50), the threaded carriage (55) including at least one positive stop (57) which limits the proximal and distal translation of the carriage (55) within the housing (20), and b. an actuator assembly (29) including a handle (42) operably attached to a chuck mover (43) for actuating engagement of the chuck (54) with the screw gear (53), a directional actuator (30) for actuating movement of the carriage (55) proximally or distally, and a carriage lock (31) configured to releasably engage a thread guide (32) with the threaded carriage (55) wherein engagement thereof permits proximal or distal movement of the threaded carriage (55) along the support shaft (51) to effect proximal or distal movement of the guidewire (15), wherein the differential control assembly (50) is actuatable to first drive the guidewire (15) into a target site by engagement of the carriage lock (31) with the carriage (55) to direct distal movement of the guidewire (15), the depth thereof being governed by the at least one positive stop (57), and to differentially drive a screw affixed to the surgical guidance device (10) by actuating the handle (42) to engage the chuck (54) with the screw gear (53), and wherein the

guidewire (15) is constructed from a material having sufficient column strength for penetrating a bone.

2. A surgical guidance device (10) comprising: a. a housing (20) having a proximal end (21), a center drive barrel (24), and a distal end (22), the proximal end (21) adapted with a hand grip (40) for gripping the surgical guidance device (10), the center drive barrel (24) configured to contain a differential control assembly (50) housed therein, the proximal end (21) of the housing (20) adapted for securing the housing (20) to a powered driving device (500), b. the differential control assembly (50) being cannulated for passage of a guidewire therethrough, and aligned along a common axis with and within the housing (20), the differential control assembly (50) including a support shaft (51) having a proximally positioned drive connector (59) for engagement with the powered driving device (500), and a distally positioned tool connector (52) configured for attachment of a driver tool guide (200), a screw gear (53) and a chuck (54) that are releasably engagable to direct rotational movement of the screw gear (53) when the chuck (54) is engaged therewith, a threaded carriage (55) translatable along the support shaft (51) for directing proximal or distal movement of the guidewire (15) through the differential control assembly (50), the threaded carriage (55) including at least one positive stop (57) which limits the proximal and distal translation of the carriage (55) within the housing (20), and c. an actuator assembly (29) including a hand grip (40) including a handle (42) operably attached to a chuck mover (43) for actuating engagement of the chuck (54) with the screw gear (53), a directional actuator (30) for actuating movement of the carriage (55) proximally or distally, and a carriage lock (31) configured to releasably engage a thread guide (32) with the threaded carriage (55) wherein engagement thereof permits proximal or distal movement of the threaded carriage (55) along the support shaft (51) to effect proximal or distal movement of the guidewire (15), wherein the differential control assembly (50) is actuatable to first drive the guidewire (15) into a target site by engagement of the carriage lock (31) with the carriage (55) to direct distal movement of the guidewire (15), the depth thereof being governed by the at least one positive stop (57), and to differentially drive a screw affixed to the surgical guidance device (10) by actuating the handle (42) to engage the chuck (54) with the screw gear (53), and wherein the directional actuator (30) is actuated to direct the guidewire (15) to move distally, and is released to retract the guidewire to move proximally, wherein displacement of the guidewire (15) is governed by the at least one positive stop (57), and wherein retraction of the guidewire (15) with the directional actuator (30) thereby differently permits distally directed driving of only the screw gear (53).

3. The surgical guidance device (10) according to claim 1 that includes a fixed grip (41).

4. The surgical guidance device (10) according to claim 1 wherein the driver tool (200) includes a drill guide.

5. The surgical guidance device (10) according to claim 1 wherein the differential control assembly (50) includes a connector (51) for securing to the powered driving device (500) and the powered driving device is a drill or power drill.

6. The surgical guidance device (10) according to claim 1 wherein the surgical guidance device (10) includes a trigger (70) for disengaging the screw gear (53) to stop rotation of a screw affixed to the surgical guidance device (10).

7. The surgical guidance device (10) according to claim 1 wherein surgical guidance device (10) includes a guidewire lock (71) for locking the guidewire in place to prevent one or more of its rotation, proximal translation, distal translation, or a combination thereof.

8. The surgical guidance device (10) according to claim 1 wherein the driver tool (200) is constructed to cooperate with a conjugately shaped portion of a screw and is adapted to receive passage of the guidewire therethrough.

9. The surgical guidance device (10) according to claim 1 wherein the directional actuator (30) is actuated to direct the guidewire (15) to move distally, and is released to retract the guidewire to move proximally, wherein displacement of the guidewire (15) is governed by the at least one

positive stop (57), and wherein retraction of the guidewire (15) with the directional actuator (30) thereby differently permits distally directed driving of only the screw gear (53).

10. The surgical guidance device (10) according to claim 2 wherein the guidewire (15) is constructed from a material having sufficient column strength for penetrating a bone.

11. The surgical guidance device (10) according to claim 1 wherein the surgical guidance device (10) has a fixed length from proximal to distal wherein rotation and translation of the differential control assembly (50) occurs within the housing (20).

12. The surgical guidance device (10) according to claim 1 wherein a proximal end (20) of the housing (20) is integral with the powered driving device (500).

13. The surgical guidance device (10) according to claim 1 wherein the at least one positive stop is at least one ring that interferes within the housing to limit movement of the carriage (55).

14. The surgical guidance device (10) according to claim 1 wherein at least one positive stop includes two or more rings oriented proximally and distally relative to the carriage (55).

15. The surgical guidance device (10) according to claim 1 wherein the positive stop is a sleeve that encloses the carriage (55) to thereby limit its translation.

16. A method for directing sequentially a guidewire (15) and a screw into target bone by differentially directing the placement of a guidewire into bone and thereupon preventing further displacement of the guidewire (15) into bone and thereafter introducing the screw into bone, wherein the guidewire (15) is either maintained in place within the bone, or is withdrawn before or as the screw is inserted into the bone, the method employing a surgical guidance device (10) according to claim 1.

17. The surgical guidance device (10) according to claim 2 wherein the proximal end (20) of the housing (20) is integral with the powered driving device (500).

18. The surgical guidance device (10) according to claim 2 wherein the at least one positive stop is at least one ring that interferes within the housing to limit movement of the carriage (55).

19. The surgical guidance device (10) according to claim 2 wherein at least one positive stop includes two or more rings oriented proximally and distally relative to the carriage (55).

20. The surgical guidance device (10) according to claim 2 wherein the positive stop is a sleeve that encloses the carriage (55) to thereby limit its translation.
